Scale Decoupled Pyramid for Object Detection in Aerial Images

You Ma, Lin Chai, and Lizuo Jin

***Abstract*— Object detection in aerial images is a challenging task for two main reasons: small object and scale variation. Existing methods utilize multilevel features to solve the scale variation problem but ignore the scale confusion problem of shallow features, limiting the small object detection performance. To solve this issue, we propose a scale decoupling module (SDM) to emphasize small object features by eliminating large object features in shallow layers. Moreover, a sparse nonlocal attention (SNLA) and an adaptive anchor matching strategy (AAMS) are proposed to further improve the small object detection performance. The SNLA only aggregates contextual information of specific sparse positions, which not only refines small object features but also is computationally friendly. The AAMS is suitable for the measurement of small objects, and it can assign more positive samples to small objects. Extensive experiments on three challenging aerial datasets, VisDrone-DET2019, UAVDT, and DIOR, demonstrate the effectiveness and adaptivity of our method. The code will be available online (https://github.com/MaYou1997/SDP).**

***Index Terms*— Aerial image, anchor assignment, object detection, scale decoupling, sparse nonlocal attention (SNLA).**

I. Introduction

THE UAV is characterized by simple operation, high flexibility, and wide field of vision, so it has found applications in many fields, such as intelligent transportation, safety search and rescue, and natural disaster early warning systems [1], [2], [3]. Object detection in aerial image is an indispensable part of these applications. Along with the rapid development of deep learning, many significant advances have been made in object detection technology [4], [5]. However, these existing methods mainly model for general scene datasets, such as microsoft common objects in context (MSCOCO) [6] and PASCAL visual object classes (PASCAL VOC) 2012 [7]. Compared to general scenes, aerial scenes have many difficulties, such as scale variation, small objects, complex backgrounds, and diverse viewpoints. Overall, developing effective aerial object detectors has become a crucial and challenging task.

Considering that small objects are the bottleneck of object detection performance, recent advances in object detection in aerial images have mainly focused on small object detection. Three mainstream methods are introduced here: contextual information-based methods [8], [9], feature fusion-based methods [10], [11], and anchor assignment-based methods [12], [13]. Contextual information-based methods, which improve the discriminability of small objects by capturing long-range dependencies. Nonlocal attention [14] is widely used in this method due to its ability to effectively aggregate global contextual information. However, it consumes huge computational and memory resources. In comparison, feature fusion-based methods have less computational complexity. It enhances the semantic information of small objects by fusing shallow and deep features and then uses the shallow layer to be responsible for small object detection. However, feature fusion can exacerbate the problem of scale confusion in the shallow layer, i.e., the shallow layer contains both large and small object features. Take the shallow feature of the baseline you only look once version 5 (YOLOv5) as an example, as shown in Fig. 1. On the one hand, we find that the feature response of large objects in the shallow layer dominates, resulting in

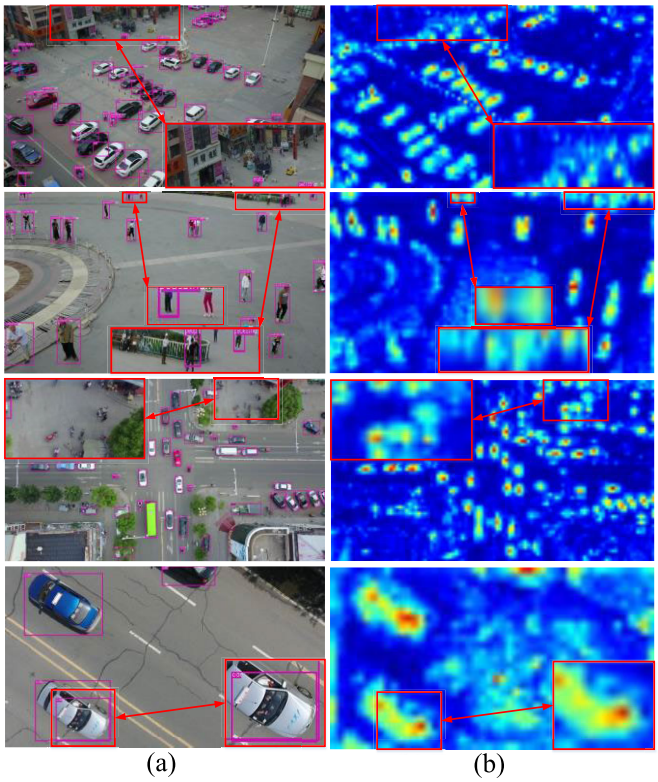


Fig. 1. Qualitative analysis of baseline YOLOv5. There are problems of missed and false detections of small objects in the baseline detection results. (a) Baseline result. (b) Baseline feature.

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small objects being missed due to weak features. On the other hand, the local high response of large objects in the shallow layer causes false detection in local regions. Therefore, effective removal of large object features in the shallow layer is a problem of concern. The above two methods mainly improve the small object detection performance by enhancing the small object feature representation. Unlike them, the anchor assignment-based method, which improves small object performance by matching more positive samples to small objects. This method has received wide attention since it does not impose additional computational consumption during the inference process. Unfortunately, existing anchor assignment methods are mostly based on the IoU metric to measure the similarity between two boxes, which is characterized by scale imbalance, i.e., large objects can assign more positive samples than small objects. In general, the performance of object detection in aerial images can be effectively improved if the shallow small object features are enhanced, while the large object features are suppressed as well as the anchor assignment metrics with a scale balance characteristics being designed.

In this article, we propose a scale decoupled pyramid method for solving the above problem. It consists of two core components: a scale decoupling module (SDM) and an adaptive anchor matching strategy (AAMS). Its overall functions include the following. First, SDM fuses feature fusion and attention mechanisms to refine the object feature representation. Then, it further enhances small object features by eliminating large objects in the shallow layer, i.e., only small object features are retained in the shallow layer. In particular, using our proposed sparse nonlocal attention (SNLA) refinement features, it achieves comparable performance to nonlocal attention (NLA) but significantly reduces the computational and memory complexity. Second, in the training phase, AAMS performs anchor assignment on the output features of the SDM. In AAMS, it uses a newly proposed metric based on position and shape similarity (PSS) to measure the distance between two bounding boxes. Compared to IoU, the PSS of large and small objects has the same sensitivity to position shifts. Therefore, the AAMS can better stimulate the potential of existing detectors for small object detection tasks. To sum up, the contributions are fourfold as follows.

1) We propose an SNLA that adaptively captures specific sparse position contextual information to enhance small object feature representation.
2) We develop an AAMS with scale balance characteristics that can assign more positive samples to small objects.
3) We propose a scaled decoupling module, which can alleviate the problem of small object miss and false detections.
4) Extensive experiments show that our proposed scale decoupled pyramid method achieves remarkable detection results on the challenging VisDrone2019-DET [15], UAVDT [16], and DIOR [17] benchmarks.

II. Related Work

A. General Object Detection

Currently, general object detectors can be divided into two categories: one-stage networks and two-stage networks. One-stage networks aim to improve the detection speed by directly predicting objects, such as single shot multibox detector (SSD) [18] and you only look once (YOLO) series [19], [20], [21]. In contrast, two-stage networks consist of two steps: first generating candidate region proposals and then refining the candidate region proposals to obtain accurate object categories and precise locations, which leads to high accuracy but inefficient, such as fast region-CNN (R-CNN) [22], faster R-CNN [23], and masked R-CNN [24]. Although existing object detection methods have superior performance in general scenarios, they fail to achieve satisfactory results when directly used for aerial scenarios. The main reason is that there is a shift in the distribution of object scales in the two domains.

B. Object Detection in Aerial Images

Benefiting from the availability of a large number of aerial images, aerial detectors, such as Oriented RCNN [25], anchor-free oriented proposal generator (AOPG) [26], dual-aligned oriented detector (DODet) [27], and general gaussian heatmap label assignment (GGHL) [28], have been developed. Recently, the main obstacle that hinders the performance of object detection in aerial images is small object. Given that small objects are mainly concentrated in the shallow layer of the feature pyramid, but the semantic information in the shallow layer is insufficient, an effective solution is to enhance the feature representation of small objects in the shallow layer using contextual information. Nonlocal attention [14] is widely used by virtue of its superior ability to capture global information. Huang et al. [29] proposed a novel nonlocal-aware pyramid attention-guided network focusing on effective features and suppressing background noise. Li et al. [30] proposed a solo-to-collaborative dual attention network to enhance shallow small object feature representation. Unfortunately, the computationally unfriendly mechanism of NLA severely limits its application scenarios. Therefore, it remains to be explored how to improve the efficiency of the attention mechanism.

In order to reduce the introduction of additional computational effort in the baseline model, some literature explores feature fusion-based methods to refine small object features. Sun et al. [31] proposed a feature refinement network, which integrates multiscale features through a bidirectional feature aggregation module. Zhang et al. [8] designed a composite semantic feature fusion structure to generate effective semantic descriptions for shallow layers. While this method is simple and effective, the fusion of deep and shallow features is accompanied by the transfer of large object features to the shallow layer, leading to the weakening of small object features. Unlike that method, we add an SDM after feature fusion to alleviate the scale confusion problem by suppressing large object features in the shallow layer.

The above two methods improve the performance of small object detection from the perspective of optimizing discriminative features, and some reports solve the problem from the perspective of anchor assignment. Duan et al. [32] developed an anchor matching strategy based on center point translation to assign more positive samples to small objects. Yang and Cai [33] proposed a group assignment strategy, which first groups according to the IoU metric between anchors and

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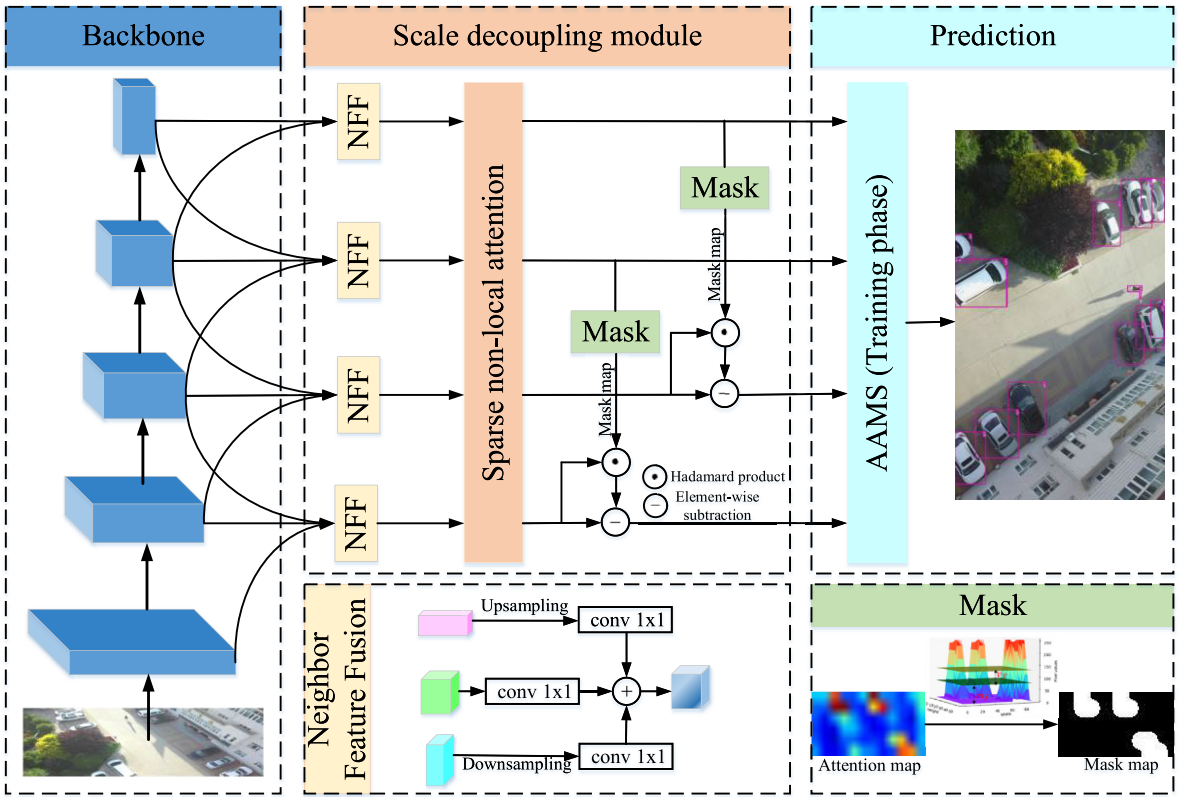


Fig. 2. Framework of our method. We first input the features extracted by the backbone network into the SDM to achieve feature enhancement and scale separation. Next, anchors are assigned using AAMS during training. Finally, the instance-level prediction results are output.

ground truth (GT) and then adaptively assigns positive or negative sample labels to each group of anchors. Although existing anchor assignment methods achieve promising results, they all assign anchors using IoU metrics that are not suitable for small objects. To this end, we focus on designing a more reasonable metric to replace the IoU metric.

III. Proposed Method

In this section, we detail the scale decoupled pyramid method. We first describe the framework overview of the method and then introduce the implementation of our proposed SNLA, SDM, and adaptive anchor assignment strategy.

A. Framework Overview

As shown in Fig. 2, we use scale decoupled pyramids for object detection in aerial images. It mainly includes an SDM and an AAMS. The SDM can generate scale-aware features, i.e., deep layers only contain large object features, and shallow layers only contain small object features. The AAMS has the advantage of scale balance, which means that using AAMS instead of IoU-based strategy can assign more positive samples to small objects.

B. SNLA

Recently, NLA [14] demonstrated that capturing all position context information in feature maps can effectively improve the object detection performance. It can rapidly capture long-range dependencies by computing the relationship between any two positions in the feature map. This article uses an embedded Gaussian by default to calculate the relationship between positions. We are given an input feature map $X \in R^{C\times H\times W}$, where C, H, and W represent the number of channels, height, and width, respectively. The NLA is shown in Fig. 3(a), which can be described as

$$A = \text{softmax}(\theta^T(X)\phi(X)) \tag{1}$$

$$Y = Ag(X) \tag{2}$$

$$Z = X + \varphi(Y) \tag{3}$$

where $\theta(\cdot)$, $\phi(\cdot)$, $g(\cdot)$, and $\varphi(\cdot)$ all denote a 1×1 convolution operation and $A \in R^{HW\times HW}$ denotes a dense affinity matrix.

Small objects are detected in high-resolution shallow features. From (1), it can be known that calculating the shallow affinity matrix A requires huge memory and computing resources. To compensate for the lack of NLA, an SNLA is proposed, which can significantly reduce the computational cost by capturing contextual information of specific sparse positions. The main rationale for the sparse design is: 1) the high and low responses of the region in the feature map generally reflect the importance of the information contained in the region and 2) most of the regions in the feature map obtained by the backbone get very low-response values. Intuitively, it is a feasible solution to use high-response positions to replace all positions for contextual information aggregation. Compared with NLA, the core change of SNLA [Fig. 3(b)] is to introduce a sparse position filter (SPF), which can adaptively select $N \ll HW$ (dynamic changes) positions with high response in the feature map.

The SPF is shown in Fig. 3(c). Specifically, we first average the input feature map $X \in R^{C\times H\times W}$ according to the channel

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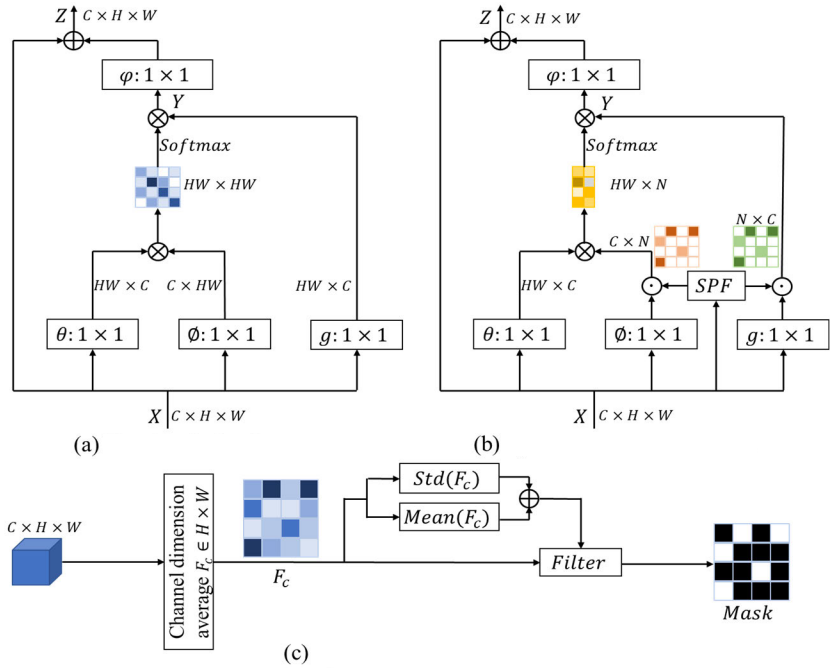


Fig. 3. Comparison of NLA and SNLA. NLA aggregates contextual information at all positions, while SNLA only considers the relationship between high-response positions. (a) NLA. (b) SNLA. (c) SPF.

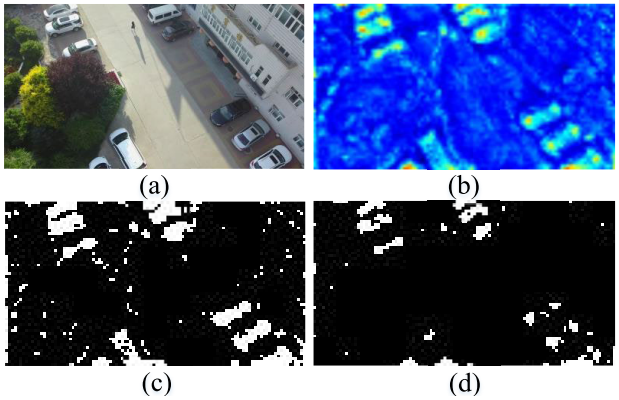


Fig. 4. Visualization of SPF. (a) Original image. (b) Feature map. (c) Filtered with m. (d) Filtered with the sum of m and v.

dimension to obtain the heatmap F_c [Fig. 4(b)], and the formula is given as follows:

$$F_c(i,j) = \frac{1}{C}\sum_{k=1}^{C} X(k,i,j)(i = 1,\ldots,Hj = 1,\ldots,W). \tag{4}$$

Then, calculate the mean m and standard deviation v of F_c. m reflects the overall response level of F_c and a high m indicates a high response in most positions and vice versa. v reflects which positions of F_c have high response and a high v indicates that high response is concentrated in local positions. Finally, filtering F_c using the sum of m and v as a threshold can adaptively select high-response positions. The SPF operation is visualized in Fig. 4. In Fig. 4(d), the sum of m and v is used to filter F_c, leaving only a few regions of high response in the mask map. The complexity of SNLA to obtain the density matrix is $HWNC(N \ll HW)$, while that of NLA is $(HW)^2C$. In other words, SPL significantly reduces the computational complexity of the attention module. The implementation of SPF is summarized in Algorithm 1.

Algorithm 1 Sparse Position Filter

Input: Feature map $X \in R^{C\times H\times W}$
Output: Mask map M

```
Calculate F_c ∈ R^{H×W} by Eq (4)
Calculate the mean of F_c: m = Mean(F_c)
Calculate the standard deviation of F_c: v = Std(F_c)
Calculate response threshold: t = m + v
for i to H do
    for j to W do
        if F_c(i, j) ≥ t then
            M(i, j) = 1
        else
            M(i, j) = 0
        end if
    end for
end for
```

The SNLA can be described as

$$Y = \text{softmax}(\theta^T S(\phi))S(g) \tag{5}$$

$$Z = X + \varphi(Y) \tag{6}$$

where $S(\cdot) \in R^{C\times N}$ denotes the sparse operation.

C. SDM

Small objects are the bottleneck restricting the performance of object detection in aerial images. It is mainly distributed in the shallow layers of the feature pyramid, but the shallow

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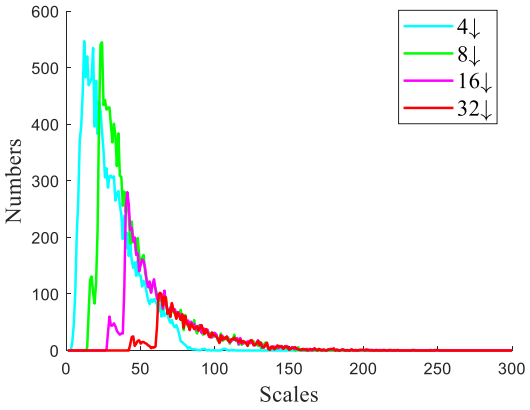


Fig. 5. Statistical scale distribution of detection objects in different feature layers in the VisDrone-DET2019 dataset.

features have scale confusion problems. We observe two phenomena: 1) the feature response of large objects in the shallow layer dominates, which leads to the missed detection of small objects, and 2) the local region of the large object in the shallow layer has a high response and the anchor scale assigned by the shallow layer is small, thus causing the false detection of the local region of the large object. How to suppress the large object features in the shallow layer is a question worth exploring. Intuitively, the small object information gradually disappears as the network becomes deeper, which means that the deep layer does not contain small object features. The range of scale processing for different downsampling feature layers is shown in Fig. 5. On the one hand, considering that the scale distribution of nonneighbor feature layers is obviously different, a skip feature subtraction structure is constructed to erase large object features in shallow layers. On the other hand, scale overlap makes the object features in neighbor feature layers complementary. To this end, we design a neighbor feature fusion (NFF) module to enhance the features.

We define the downsampling feature layers of $2\times$, $4\times$, $8\times$, $16\times$, and $32\times$, respectively, as p_2, p_4, p_8, p_{16}, and p_{32}. The SDM is shown in Fig. 2, and it consists of four steps.

Step 1: The feature representation is enhanced by NFF

$$NFF(p_{2^i}) = \begin{cases} h(p_{2^{i-1}}) + \mathcal{C}(p_{2^i}) + f(p_{2^{i+1}}), & i = 2, 3, 4 \\ h(p_{2^{i-1}}) + \mathcal{C}(p_{2^i}), & i = 5 \end{cases} \tag{7}$$

where Maxpooling 2×2 stride $= 2$, Conv1$\times$1stride $= 1C_{\text{out}} = 2C_{in}$, $\mathcal{C}(\cdot)$: Conv1 $\times$ 1stride $= 1C_{\text{out}} = C_{\text{in}}$, $f(\cdot)$: upsampling $2\uparrow$, and Conv1 $\times$ 1stride $= 1C_{\text{out}} = 0.5C_{\text{in}}$.

Step 2: Refine Step 1 output features using SNLA

$$\mathcal{F}(p_{2^i}) = \text{SNLA}(\text{NFF}(p_{2^i})) \quad i = 2, 3, 4, 5. \tag{8}$$

Step 3: We first calculate the average value of $\mathcal{F}$ according to the channel dimension to obtain the attention map $\bar{\mathcal{F}}$, then calculate the average $m_{\bar{\mathcal{F}}}$ and standard deviation $v_{\bar{\mathcal{F}}}$ of $\bar{\mathcal{F}}$, and finally filter $\bar{\mathcal{F}}$ using the sum of $m_{\bar{\mathcal{F}}}$ and $v_{\bar{\mathcal{F}}}$ to obtain the mask map

$$\text{Mask}(i, j) = \begin{cases} 1, & \bar{\mathcal{F}}(i, j) \geq m_{\bar{\mathcal{F}}} + v_{\bar{\mathcal{F}}} \\ 0, & \bar{\mathcal{F}}(i, j) < m_{\bar{\mathcal{F}}} + v_{\bar{\mathcal{F}}}. \end{cases} \tag{9}$$

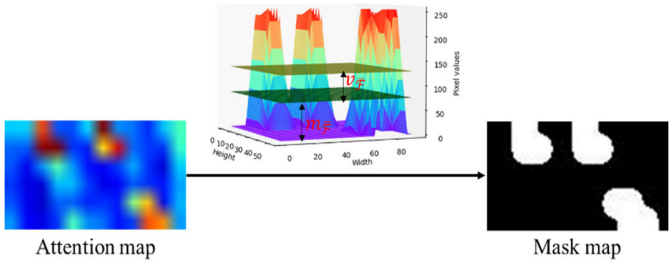


Fig. 6. Generation of mask map. High-response regions are retained and low-response regions are suppressed.

The generation of the mask map is visualized from a 3-D perspective in Fig. 6. Using the sum of $m_{\bar{\mathcal{F}}}$ and $v_{\bar{\mathcal{F}}}$ can effectively extract high-response regions in the attention map. In general, large objects in deep feature layers respond strongly, so implementing Step 3 can separate large objects from the deep layer.

Step 4: Build skip structures to eliminate large object features in shallow feature layers

$$\text{SDM}(p_{2^i}) = p_{2^i} \ominus (p_{2^i} \odot up(\text{Mask}(p_{2^{i+2}}))) i = 2, 3 \tag{10}$$

where $\ominus$ denotes element-wise subtraction, $\odot$ denotes Hadamard product, and $up(\cdot)$ denotes upsampling $4\times$.

D. AAMS

ATSS points out the importance of positive and negative sample assignment, and designing a reasonable assignment strategy can significantly affect the model performance [34]. Existing IoU-based anchor assignment methods have three shortcomings: 1) the sensitivity of small objects and large objects to position deviation is different; 2) the shape similarity between anchor and ground truth is not considered; and 3) the information difference in different regions of the ground truth is not distinguished. Since the anchor obtains the predicted box corresponding to the ground truth through translation and scaling operations, anchors with a shape similar to the ground truth are more suitable for regression task training. In addition, most of the bounding regions of ground truth contain background, so the amount of information contained in the central region of ground truth is usually stronger than that of bounding regions.

Inspired by the above ideas, we propose an anchor localization quality evaluation metric based on PSS. We define G_x, G_y, G_w, and G_h as the center coordinates, height, and width of the ground truth G, respectively; A_x, A_y, A_w, and A_h as the center coordinates, height, and width of the anchor A, respectively. PSS is defined as follows:

$$\text{PSS}(G, A) = e^{-\frac{L_2([G_x, G_y, \frac{G_w}{2}, \frac{G_h}{2}], [A_x, A_y, \frac{A_w}{2}, \frac{A_h}{2}])}{T}} \tag{11}$$

where $L_2(\cdot, \cdot)$ represents the calculation of the L2 distance between the two bounding boxes and T represents the temperature coefficient.

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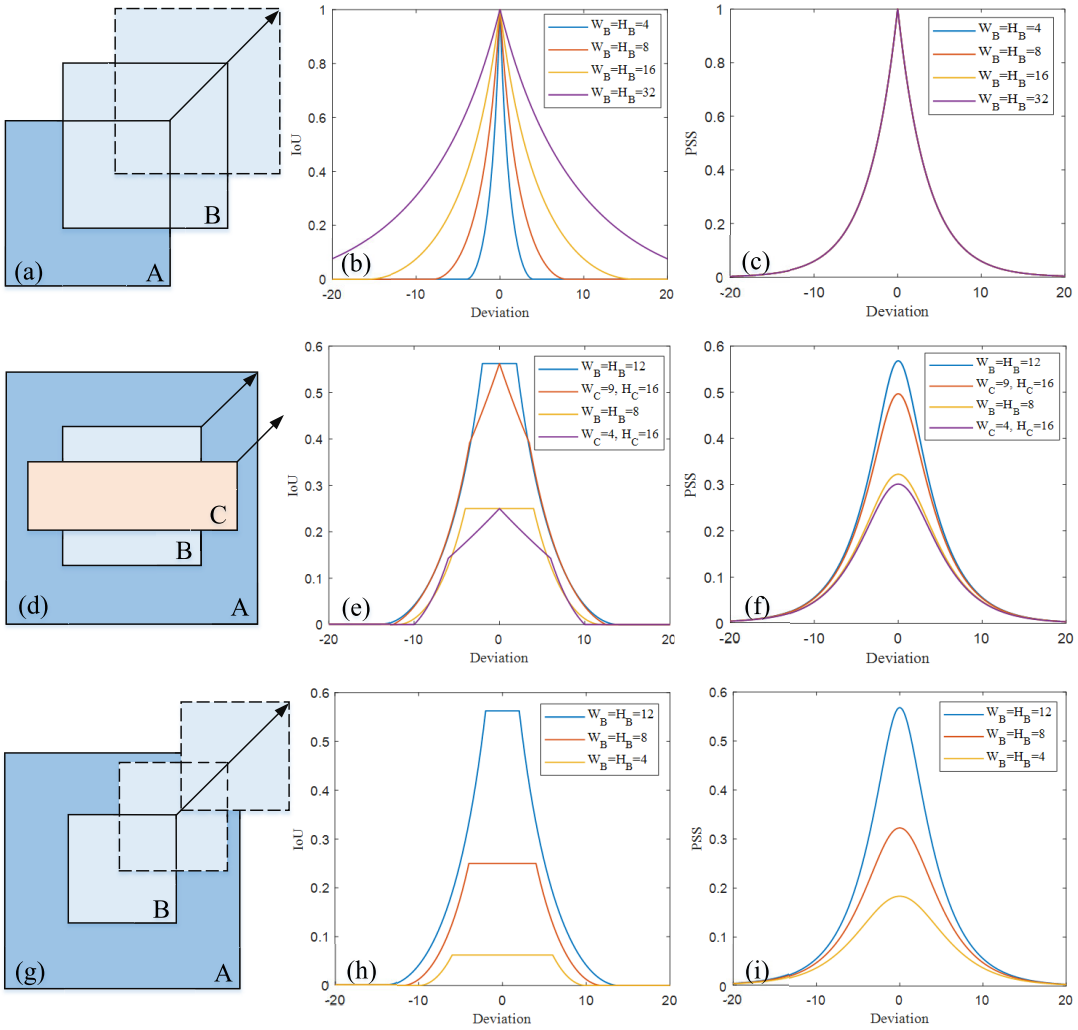


Fig. 7. Comparison between the IoU-deviation curve and the PSS-deviation curve in different scenarios. The abscissa denotes the deviation between the center points of the two coordinate boxes, and the ordinate denotes the value of the metric. (a) Scenario 1: A and B have the same size. (d) Scenario 2: A and B have the same shape but different scales, and B and C have the same area but different shapes. (g) Scenario 3: the scale of A is fixed, and the scale of B varies. (b), (e), and (h) IoU-deviation curves for different scenarios, respectively. (c), (f), and (i) PSS-deviation curves for different scenarios, respectively.

TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE VISDRONE-DET2019 DATASET

Methods	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Model Size (MB)
Cascade-RCNN[37]	23.2	39.9	23.4	16.5	36.8	39.4	273.2
YOLOv5[21]	24.1	44.1	24.7	15.3	35.6	38.4	93.0
RetinaNet[5]	24.3	44.3	18.7	18.7	35.2	37.8	251.7
Libra RCNN[38]	24.3	41.2	24.9	16.8	34.0	36.8	185.4
HawkNet[39]	25.6	44.3	25.8	19.9	36.0	39.1	130.9
VFNet[40]	25.9	42.1	27.0	16.8	37.3	41.4	296.2
DetectoRS*[41]	26.8	43.2	28.0	17.5	38.2	41.7	540.1
DMNet[42]	28.2	47.6	28.9	19.9	39.6	**55.8**	-
ClusDet[43]	28.4	53.2	26.4	19.1	**40.8**	54.4	-
RRNet[44]	29.1	**55.8**	27.2	-	-	-	-
Ours	**30.2**	52.5	**30.6**	**22.6**	39.6	39.8	96.7

In Fig. 7, the differences between the IoU-deviation curves and the PSS-deviation curves under the three scenarios are analyzed.

Scenario 1: In Fig. 7(a), A and B are the same size. When B deviates from A along the arrow direction, it can be seen from the IoU-deviation curve [Fig. 7(b)] that the small object is more sensitive to the deviation of position than the large object. However, the PSS-deviation curves of different scale objects overlap [Fig. 7(c)], indicating that PSS is scale insensitive.

Scenario 2: In Fig. 7(d), A and B have the same shape but different scales, and B and C have the same area but different shapes. Comparing Fig. 7(e) and (f), when B and C

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are included in A, the IoU of B and C are the same, but the PSS of B is higher than that of C. Given that B is more similar to A than C, PSS reflects the importance of shape similarity better than IoU.

Scenario 3: In Fig. 7(g), the scale of A is fixed, and the scale of B varies. Comparing Fig. 7(h) and (i), when B is inside A, the IoU at different positions is the same, but the PSS is different, and the PSS near the center of A is higher. This demonstrates that PSS can reflect the position difference of anchors.

Object detection is a joint task of classification and localization, so a reasonable sample assignment strategy should consider the localization quality and classification quality of the anchor. We first compute the cost between ground truth g_i and anchor a_j pair-wise using a method similar to simplify optimal transport assignment (SimOTA) [35]

$$C_{ij} = L_{ij}^{cls} + \lambda L_{ij}^{reg} \tag{12}$$

where L_{ij}^{cls} and L_{ij}^{reg} denote the classification loss and regression loss, respectively. L_{ij}^{cls} uses the binary cross-entropy loss, and $L_{ij}^{\text{reg}} = 1 - \text{PSS}(g_i, a_j)$. λ denotes the tradeoff parameter. Then, the k anchors with the lowest cost are selected as positive samples for each ground truth. The process of AAMS is shown in Algorithm 2.

Algorithm 2 Adaptive Anchor Matching Strategy

Input:
$\mathcal{G}$ is a set of ground truths
$\mathcal{A}$ is a set of all anchors
Output:
$\mathcal{P}$ is a set of positive samples
$\mathcal{N}$ is a set of negative samples
for each ground truth $g \in \mathcal{G}$ do
  Build an empty positive sample set for ground truth g: $\mathcal{P}_g \leftarrow \emptyset$
  for each anchor $a \in \mathcal{A}$ do
    Calculate the PSS between $a$and g by Eq. (11)
    Calculate the cost between a and g by Eq. (12)
  $\mathcal{P}_g \leftarrow$ select k anchors from $\mathcal{A}$ that have the lowest cost with ground truth g
  $\mathcal{P} = \mathcal{P} \cup \mathcal{P}_g$
  end for
end for
When the same anchor is assigned to multiple ground truth, select the ground truth with the lowest cost.
$\mathcal{N} = \mathcal{A} - \mathcal{P}$
return $\mathcal{P}, \mathcal{N}$

IV. Experiments

In this section, we first describe the experimental setting. Next, the performance of our method is compared with the existing state-of-the-art methods on three datasets. Then, the ablation experiments were designed to analyze the contributions of different modules. To further explore the effectiveness of the module, we provide more details and comparative results. Finally, we qualitatively analyze the detection results of the model. All experiments were performed on a GeForce RTX 3090.

A. Experimental Setting

1) Datasets: VisDrone-DET2019 [15] has a total of 8599 images, including 6471 images in the training set, 548 images in the validation set, and 1580 images in the test set. The dataset contains ten classes: pedestrian, people, bicycle, car, van, truck, tricycle, awning tricycle, bus, and motor.

UAVDT [16] consists of aerial images selected from 10-h-long videos. These video sequences are collected at different locations in the city, and it covers diverse scene variations (e.g., weather, viewpoint, and illumination). The training set and test set contain 23 829 images and 16 580 images, respectively, with an image resolution of 1024 × 540. The dataset contains three categories: car, bus, and truck.

DIOR [17] is a large dataset used to evaluate object detectors in the field of remote sensing, including 23 463 images and 190 288 objects, covering 20 object categories. The DIOR dataset has a wide range of objects (scale variation).

2) Implementation Details: The stochastic gradient descent function (SGD) is employed as the loss optimization function in the experiments, the momentum is set to 0.9, the weight decay is set to 0.0005, the initial learning rate is set to 0.01, and the cosine annealing method [36] is used to adjust the learning rate. In the training process, the foreground–background class imbalance problem in the object detector is solved using the focus loss [5], and a multiscale image input strategy is utilized. The training parameter epoch is set to 120, the batch size is 16, and the IoU threshold of non-maximum suppresion (NMS) is 0.5. During the training process, data augmentation (translation, scale, and rotation transformation) is used to expand the number of training samples. In the experiment, all models are implemented by PyTorch 1.10.0.

3) Evaluation Metrics: We evaluate the detection results of our method using the average precision (AP) at different IoU thresholds. Specifically, AP_{50} indicates that the IoU threshold defined as true positive (TP) samples is 0.5. AP represents the mean value from AP_{50} to AP_{95} with an IoU interval of 0.05. Moreover, we use AP_S, AP_M, and AP_L to evaluate the detection performance of small objects, medium objects, and large objects, respectively. In terms of model inference speed, we use the FPS metric for evaluation.

B. Comparisons With State-of-the-Art Methods

Results on the VisDrone-DET2019 dataset are shown in Table I. Compared with the existing superior aerial detectors, including DetectoRS ∗ [41], DMNet [42], clustered detection (ClusDet) [43], and RRNet [44], the AP of our method is improved by 3.4%, 2.0%, 1.8%, and 1.1%, respectively. In particular, HawkNet [39] and our proposed method use 800 × 1333 and 800 × 800 resolution images for model evaluation, respectively, but the AP performance of our method is 4.6% higher than that of HawkNet [39], reflecting that our detector has stronger feature representation. The core idea of ClusDet [43] is to enlarge small objects to improve the performance of small objects; RRNet [44] expands the number of objects in the image by a resampling method to improve the object

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TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE UAVDT DATASET

Model	Backbone	Input	AP	Model Size (MB)
SSD[18]	VGG-16	512×512	33.6	206.7
Retinanet[5]	ResNet-101	512×512	33.9	251.7
SLTnet[45]	ResNeXt-101	540×1024	35.0	855.3
HawkNet[39]	ResNet-50	800×1333	37.4	131.0
LSN[46]	VGG-16	512×512	37.8	307.4
CADNet[32]	VGG-16	512×512	43.7	-
NETNet[47]	VGG-16	512×512	43.9	212.4
YOLOv5[21]	CSPDarknet53	512×512	39.3	92.8
Ours	CSPDarknet53	512×512	**45.3**	96.6

detection performance. An interesting phenomenon is that the AP_S performance of ClusDet and RRNet is inferior to our method. The benefits are mainly from the SDM and AAMS. The SDM highlights small object features by eliminating large object features in shallow layers, and the AAMS can match more positive samples for small objects. It can also be seen from Table I that our method is very efficient. Its model size is smaller than all other methods except for the baseline YOLOv5.

Results on the UAVDT dataset are shown in Table II. As can be seen from the table, our proposed method achieves the state-of-the-art performance with an AP of 45.3%. Compared with the baseline YOLOv5, the AP of our detector is improved by 6.0%. It is worth noting that even though the UAVDT dataset contains various scene images, the detection performance of our method can still achieve satisfactory results.

Results on the DIOR dataset are shown in Table III. It can be seen that our method obtains the best detection performance on nine categories. Compared with existing methods, our method gets a huge performance boost on small object categories, including harbor and vehicle, which indicates that our designed module is suitable for small object detection. The detector can achieve excellent detection performance even though the scale variation in the DIOR dataset is obvious. Although the model size of lightweight oriented object detector (LO-Det) [48] is the smallest, there is room for further improvement in its detection performance. So far, the effectiveness and applicability of our method have been fully validated.

C. Ablation Study

To evaluate the contributions of different components, we conduct detailed ablation experiments on three benchmark datasets. The results are shown in Table IV. The table shows that we compare four variants: 1) using baseline for performance evaluation; 2) introducing SDM based on variant I; 3) replacing the anchor assignment strategy in the baseline to AAMS; and 4) performance evaluation using our method.

1) Effect of SDM: Directly using baseline to evaluate aerial datasets has poor detection performance, especially for small objects. The reason is that features of various scales are mixed in the shallow layers of the baseline and the feature responses of large objects dominate, resulting in missed detection of small objects. Fortunately, the SDM obtains scale-separated features by suppressing large object features in shallow layers. In Table IV, after the introduction of the SDM module, the AP_S on the three datasets increased by 5.9%, 4.9%, and 3.2%.

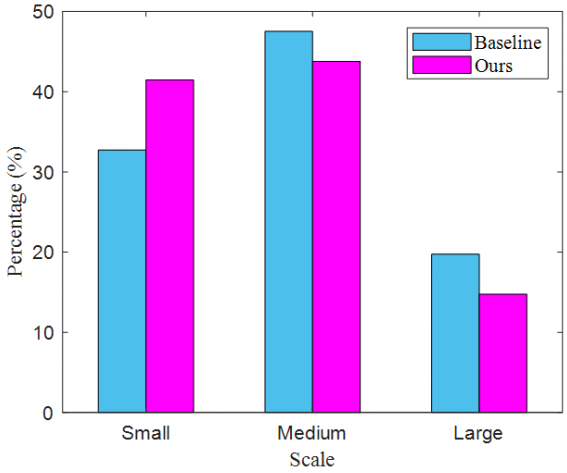


Fig. 8. Percentage of positive samples for different scale objects.

2) Effect of AAMS: The anchor assignment strategy in the baseline tends to match more positive samples to large objects, resulting in insufficient network learning of small object features. To address this problem, we improve the scale imbalance by designing an anchor assignment strategy suitable for small objects. In Table IV, after replacing the original anchor assignment strategy with AAMS, the AP_S on the three datasets increased by 2.7%, 2.5%, and 1.8%. However, AP_L decreased by 0.8%, 1.9%, and 1.4%. The main reason is that AAMS reduced the number of anchors assigned to large objects. The percentages of positive samples for objects of different scales before and after the modification of the anchor assignment strategy are shown in Fig. 8. It can be clearly seen that the proportion of positive samples assigned to small objects obviously increases after the introduction of AAMS. Although the proportion of positive samples matched by medium objects decreases, the number of positive samples assigned to a GT increases, so the medium object performance still improves slightly.

Compared with the baseline, our method achieves improvements on each evaluation metric, and the gain of evaluation metrics is shown in Fig. 9. It can be seen that

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TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR DATASET

	LO-Det [48]	YOLOv5[21]	FSoD-Net[49]	RSANet[50]	AFDet[51]	CANet[52]	ours
Backbone	MobileNetv2	CSPDarknet53	VGG-16	DLA-34	DLA-34	ResNet-101	CSPDarknet53
Airplane	72.6	81.5	**88.9**	73.6	82.4	70.3	86.8
Airport	65.0	78.6	66.9	**86.0**	81.5	82.4	81.4
Baseball field	76.7	78.5	**86.8**	72.6	81.9	72.0	83.4
Baseball court	84.7	88.4	90.2	89.6	89.8	87.8	**90.5**
Bridge	33.7	46.3	45.5	43.6	51.7	**55.7**	52.7
Chimney	73.7	73.9	79.6	75.3	74.9	**79.9**	78.1
Dam	56.8	59.4	48.2	62.3	58.7	**67.7**	63.2
Expressway service area	75.9	83.6	86.9	79.5	84.2	83.5	**87.8**
Express toll station	57.5	69.2	75.5	68.7	73.3	**77.2**	73.6
Golf course	66.3	78.8	67.0	78.6	79.5	77.3	**81.9**
Ground track field	68.0	82.2	77.3	79.1	81.0	83.6	**85.6**
Harbor	60.9	60.9	53.6	57.9	44.2	56.0	**65.2**
Overpass	51.5	57.7	59.7	59.2	62.0	63.6	**64.7**
Ship	88.6	74.0	78.3	**90.0**	77.8	81.0	80.4
Stadium	68.0	78.1	69.9	55.8	63.2	79.8	**82.6**
Storage tank	64.3	71.7	75.0	77.0	76.9	70.8	**77.8**
Tennis court	86.3	82.6	91.4	87.8	**91.0**	88.2	87.5
Train station	47.6	53.5	52.3	65.3	62.5	**67.6**	56.1
Vehicle	42.4	56.4	52.0	55.3	59.3	51.2	**59.9**
Windmill	76.7	80.8	**90.6**	86.5	87.1	89.6	84.7
Model Size (MB)	**27.0**	93.3	232.2	201.5	81.2	258.4	96.9
AP_{50}	65.9	71.8	71.8	72.2	73.2	74.3	**76.2**

TABLE IV
ABLATION STUDY RESULTS ON DIFFERENT DATASETS

	SDM	AAMS	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
VisDrone-DET2019 Dataset								
I			24.1	44.1	24.7	15.3	35.6	38.4
II	✓		28.9	50.0	29.2	21.2	39.3	40.9
III		✓	26.0	46.7	26.5	18.0	36.7	37.6
VI	✓	✓	30.2	52.5	30.6	22.6	39.6	39.8
UAVDT Dataset								
I			39.3	67.0	37.2	35.8	56.3	49.5
II	✓		43.8	72.7	41.4	40.7	60.1	51.4
III		✓	41.4	69.6	38.6	38.3	58.4	47.6
VI	✓	✓	45.3	74.5	42.7	42.5	61.1	50.3
DIOR Dataset								
I			50.4	71.8	54.7	19.1	45.2	65.7
II	✓		52.3	74.5	56.6	22.3	46.5	66.8
III		✓	51.1	73.4	55.8	20.9	45.9	64.3
VI	✓	✓	52.8	76.2	57.6	23.2	47.1	66.2

the gain of AP_S exceeds AP_M and AP_L, which is in line with the original intention of the module design to focus on the detection of small objects. In addition, the evaluation metric AP_{75} on the three datasets also increased significantly, indicating that our method can enhance the positioning quality of the regression box.

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TABLE V

COMPARING DIFFERENT FEATURE FUSION METHODS ON THE VISDRONE2019-DET DATASET. BASELINE SDM DENOTES THAT THE FEATURES EXTRACTED BY THE BACKBONE ARE DIRECTLY USED WITHOUT FEATURE FUSION

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Baseline SDM	29.3	50.6	29.5	21.9	38.2	39.1
SDM+FPN	29.6	51.3	29.8	21.7	39.3	39.7
SDM+NFF	30.2	52.5	30.6	22.6	39.6	39.8

TABLE VI

COMPARING DIFFERENT ATTENTION METHODS ON THE VISDRONE-DET2019 DATASET. H AND W REPRESENT THE LENGTH AND WIDTH OF THE INPUT FEATURE MAP, RESPECTIVELY. SNLA (MEAN) AND SNLA (MEAN + STD) REPRESENT THE FILTERING OF HIGH-RESPONSE REGIONS BY USING THE MEAN OF THE FEATURE MAPS AND THE SUM OF THE MEAN AND STANDARD DEVIATION OF THE FEATURE MAPS, RESPECTIVELY. β AND ρ REPRESENT THE PROPORTION OF HIGH-RESPONSE REGIONS IN THE FEATURE MAP, AND THERE IS A RELATIONSHIP OF $\beta < \rho < 1$. NA MEANS THAT NO ATTENTION MECHANISM IS USED

Method	AP	AP_S	AP_M	AP_L	FPS	Complexity	
						Memory	Time
+NA	29.1	21.2	38.6	39.2	23.1	0	0
+NLA[14]	30.5	23.1	39.6	40.0	19.2	$(HW)^2$	$(HW)^2$
+SNLA(Mean)	30.3	22.7	39.6	39.8	20.9	$\rho(HW)^2$	$\rho(HW)^2$
+SNLA(Mean+Std)	30.2	22.6	39.5	39.8	21.8	$\beta(HW)^2$	$\beta(HW)^2$

TABLE VII

COMPARING DIFFERENT ANCHOR ASSIGNMENT STRATEGIES ON VISDRONE-DET2019

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Baseline	24.1	44.1	24.7	15.3	35.6	38.4
+PISA[54]	24.7	44.9	25.2	16.2	36.2	38.4
+FreeAnchor[55]	25.2	45.6	25.8	16.8	36.7	38.6
+ATSS[34]	25.2	45.4	26.0	16.7	37.0	38.8
+PAA[56]	25.4	45.8	26.1	17.2	36.3	38.0
+AAMS(ours)	26.0	46.7	26.5	18.0	36.7	37.6

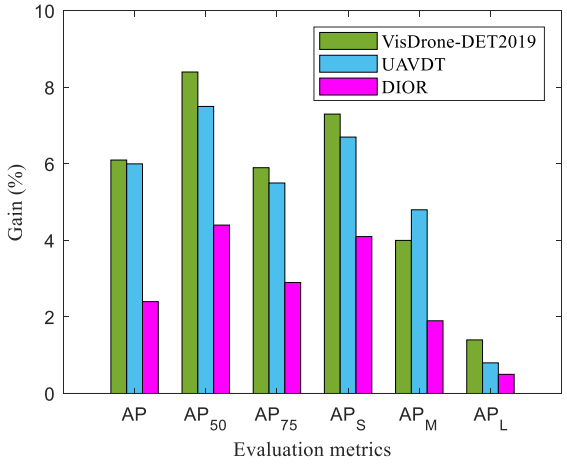


Fig. 9. Gain of evaluation metrics.

D. Module Analysis

1) Analysis of Neighbor Feature Fusion: In the SDM, feature fusion is first used to enhance feature representation. The results of our NFF and the existing feature pyramid network (FPN) [53] method are shown in Table V. The AP performance gain of 0.3% is obtained by introducing FPN in the baseline SDM. However, AP_S decreased slightly (from 21.9% to 21.7%). We hypothesize that this is because the use of FPN for feature fusion exacerbates the degree of scale confusion in the shallow features. Compared with FPN, AP can get further improvement after using NFF (29.6% versus 30.2%), and AP_S, AP_M, and AP_L are all positive gains. This suggests that NFF can mitigate scale confusion while enriching feature representation.

2) Analysis of Sparse Nonlocal Attention: Nonlocal attention can effectively improve the object detection performance. However, its computational complexity is huge, so the application scenarios are limited. The memory and time of using SNLA and NLA, respectively, are shown in Table VI. It can be seen that the AP of SNLA is slightly lower than that of NLA. The main reason is that some key information is filtered in the sparse operation of SNLA. However, the detection speed of the model using SNLA is 13.5% higher than that of the model using NLA. More interestingly, the mean of feature maps and the sum of mean and standard deviation of feature maps are used for position filtering in SNLA, respectively, but there is almost no difference in detection performance between the two

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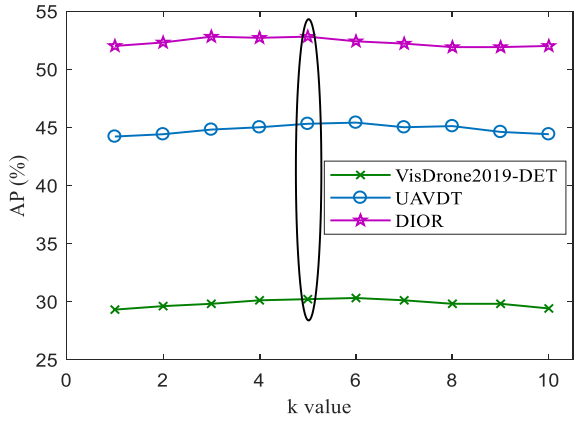


Fig. 10. Sensitivity of k value.

methods. Experimental results demonstrate the effectiveness and lightness of SNLA.

3) Analysis of Adaptive Anchor Matching Strategy: The results for different anchor assignment strategies are shown in Table VII. It can be seen that the AAMS achieves the state-of-the-art performance. The main reason for its success is the ability to treat large and small objects fairly during the anchor assignment process. What is more, the small object gets an impressive performance improvement, i.e., AP_S achieves a gain of 2.7% (from 15.3% to 18.0%). Considering the large proportion of small objects in aerial images, AAMS is suitable for object detection in aerial images.

4) Analysis of Parameter Sensitivity: The AAMS assigns k positive samples to each ground truth during implementation, where the value of k is a hyperparameter. In order to verify the influence of the k value on the performance of the model, comparative experiments are carried out by setting different k values. The results of our method on three datasets VisDrone2019-DET, UAVDT, and DIOR under different k values are shown in Fig. 10. In Fig. 10, we observe a phenomenon that the detection performance of the model on the three datasets roughly tends to increase and then decrease with the continuous increase of k. The possible reasons are given as follows. Too small k will lead to insufficient positive samples for ground truth matching, while too large k will introduce many low-quality positive samples. When $k = 5$, the sum of AP values of our method on the three datasets reaches the maximum, i.e., the best overall performance for the three tasks, so k is set to 5.

5) Analysis of Detection Error: The detection results of some samples are shown in Fig. 11. In Fig. 11(a), the results of the baseline detection suffer from missed detection and false detections, especially for small objects. This is mainly due to scale confusion in shallow layers. As can be seen in Fig. 11(b), our method can mitigate the above two deficiencies. The main benefit comes from the scale-separation features generated by the SDM.

In order to quantitatively analyze the problem of missed detection and false detection, we count the detection results of 30k objects in the VisDrone-DET2019 dataset, as shown in Fig. 12. False detection IoU < 0.5 between the predicted box and the ground truth. It can be directly seen from the figure

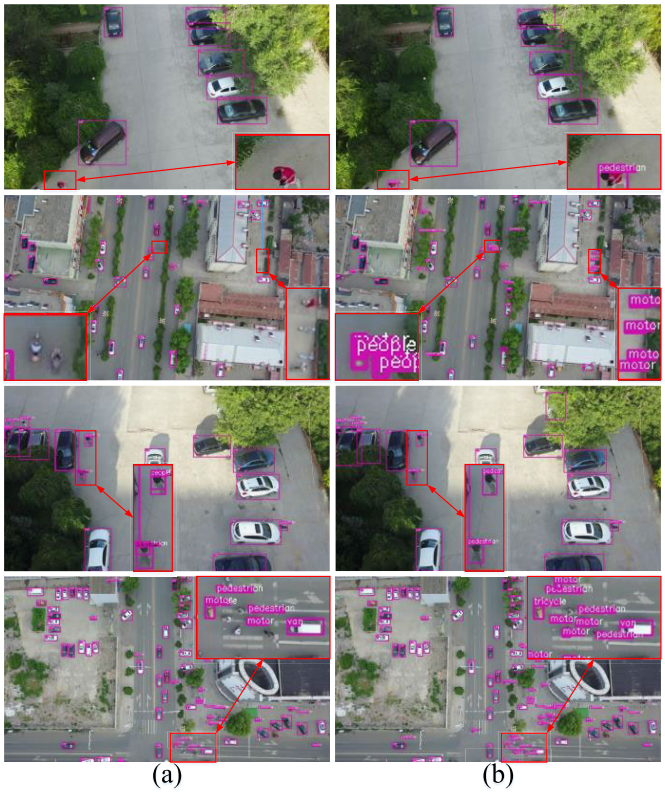


Fig. 11. Detection results. (a) Results of the baseline. (b) Results of our method.

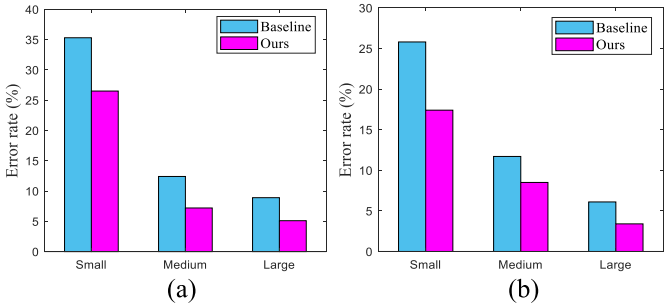


Fig. 12. Error analysis of detection results. (a) Missed detections. (b) False detections.

that our method can significantly reduce missed detections and false detections in small objects.

E. Qualitative Analysis

1) Feature Visualization: To further illustrate the effectiveness of our method, we visualize the features of the intermediate layer. The results are shown in Fig. 13. In Fig. 13(c), the response of large objects in the shallow layer is higher than that of small objects (the warmer the color, the higher the response). This phenomenon explains the low performance of small object detection. The scale decoupling features are shown in Fig. 13(d), large object features in shallow layers are erased.

2) Visualization of Detection Results: To more directly reflect the effect of our method, the detection results on different datasets are shown in Fig. 14. It can be noted that the detector can achieve high-performance object detection in

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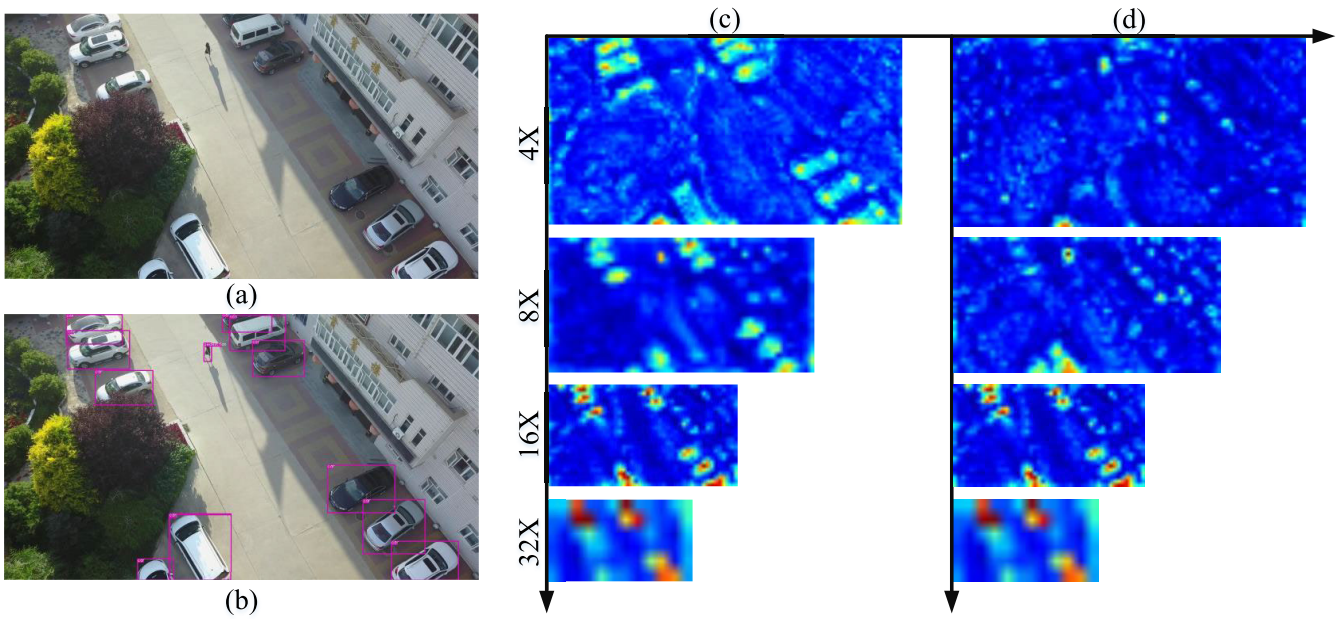


Fig. 13. Feature visualization. (a) Input image. (b) Detection results of our method. (c) Visualization results of the backbone. (d) Visualization results of our method.



Fig. 14. Detection results of different datasets. (a) VisDrone-DET2019 dataset. (b) UAVDT dataset. (c) DIOR dataset.

complex scenes such as cross-time (including daytime, dusk, and night), cross-view (including side view, front view, and top view) and cross-weather (including sunny, rainy, and foggy). In Fig. 14(c), the scale variation between different classes in the DIOR dataset is obvious (e.g., the scale difference between expressway service area and vehicle is hundreds of times), but our method can still achieve satisfactory detection results. Unfortunately, our proposed method falls short in object detection in dense scenes. The main reasons are given as follows: 1) when the detector processes dense objects of the same category, it is prone to recognize them as a whole and 2) for close objects of the same class, their positive anchors overlap severely. However, an anchor can only match one ground truth, so it is prone to the problem of unbalanced sample matching.

V. Conclusion

In this article, we propose a scale decoupled pyramid network for object detection in aerial image. By introducing an SDM, the large object features in the shallow layers can be eliminated to highlight the small object features. Specifically, this module includes a computationally friendly SNLA that refines small object feature representations by adaptively aggregating contextual information from high-response locations in the feature map. Besides, we develop a simple but effective AAMS to assign more positive samples to small

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objects. Extensive experimental results on three challenging aerial datasets demonstrate the effectiveness of our method, especially for small object detection. Although our method clearly improves the detection performance of small objects, there is still room for improvement. In the future, we will further explore the advantages of scale decoupled pyramid to extend to the video domain with temporal information.

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